

SOC4190 BLOCKCHAIN BASICS & PLATFORMS

Wallets, Forks and Types of Blockchain

Mode of teaching: **Online, asynchronous**

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Crypto giant Binance hit by $6.3bn hack

My video

What is Binance?

The world's largest cryptocurrency exchange, serving as a primary gateway to the global digital economy.



How size of Binance crypto hack compares

Some of the largest known hacks since Bitcoin launch in 2008



$6.3B
Estimated total funds stolen.

$3.0B
Funds frozen on-chain.

Bitcoin (BTC)
Drops to six-week low below $77k.

Binance (BNB)
Falls 5% to trade near $709.

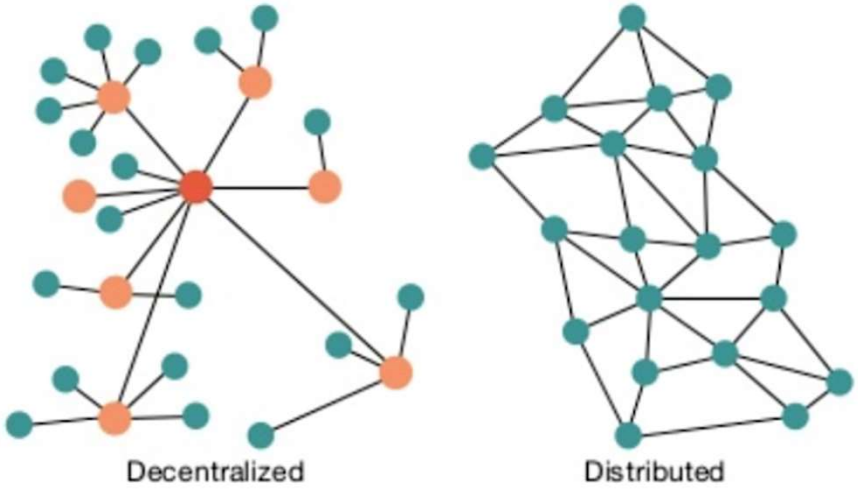
Crypto Stocks
Major equities decline nearly 5%.

Who Controls What

Stop	Today's question	What we know	
The place	Where should a key live?	—	?
The keys	Can one key be too few?	—	?
The rules	How do the rules of a blockchain change?	—	?
The split	What happens to my coins when a chain splits?	—	?
The members	Who may read, write and join?	—	?
The agreement	Can a network agree without mining?	—	?

My video

What happens when the rules themselves change?



A blockchain has no boss. So who decides where your key is kept, who may change the rules, and who may take part at all? Six questions — the last lecture of the Bitcoin half of the course.

No boss: so who controls the key, the rules and the membership?

Where We Stopped

A Story: Three Places for Your Savings

My video

Lecture 5: a wallet keeps the key, and the key is the money. So the old question about savings — pocket, safe or bank — becomes a question about the key.

Each place trades **convenience** against **safety**, and the bank adds a third thing: **trust in someone else**.

IN YOUR POCKET

ready to spend in a second

lost or stolen in a second

IN A SAFE AT HOME

slow to reach

safe from pickpockets

IN A BANK

the bank keeps it for you

only as safe as the bank

Aziz has savings. Where he keeps them decides how easy they are to use — and to lose.

Where you keep the key is a trade between convenience, safety and trust.

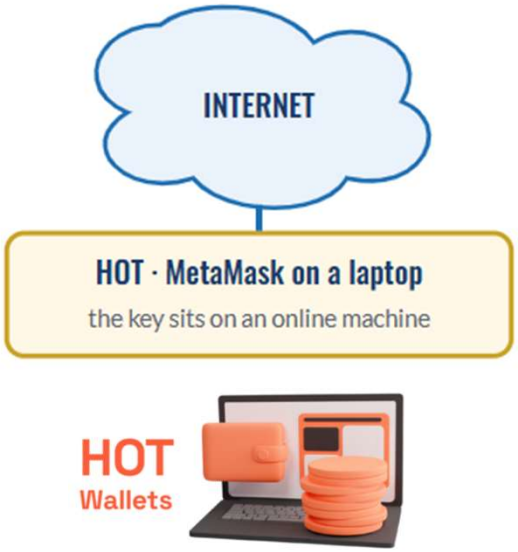
Keep this picture

The next slides give each of the three places its wallet name.

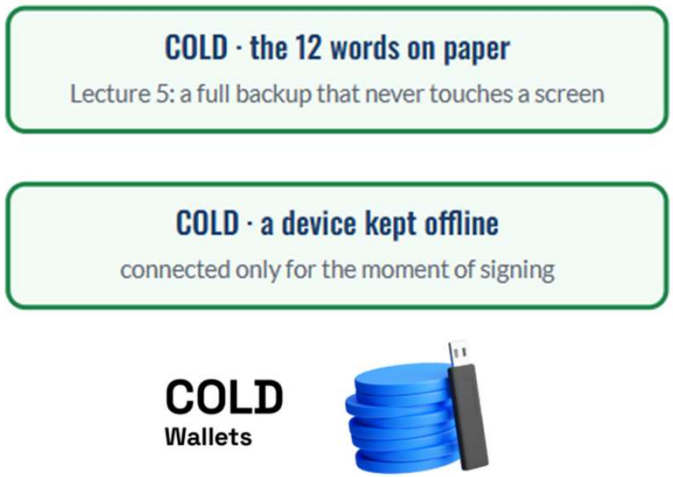


A Story: Three Places for Your Savings

My video



no connection



New word · Hot wallet

A wallet whose key sits on a device connected to the internet — a browser extension, a phone app. Quick to use; open to anything that reaches that device.

New word · Cold wallet

A key kept where the internet cannot reach it: on paper, or on a device that stays offline. Slow to use; safe from remote thieves.

Hot for spending, cold for saving.

Your course wallet is hot — right for free test coins. People keep large savings cold, and a small spending amount hot, like a safe and a pocket.

New Word: Hardware Wallet

New word · Hardware wallet

A small device made only to keep a private key and sign with it. The computer sends it a payment; it shows the payment on its own screen; you press its button; it sends back only the signature.

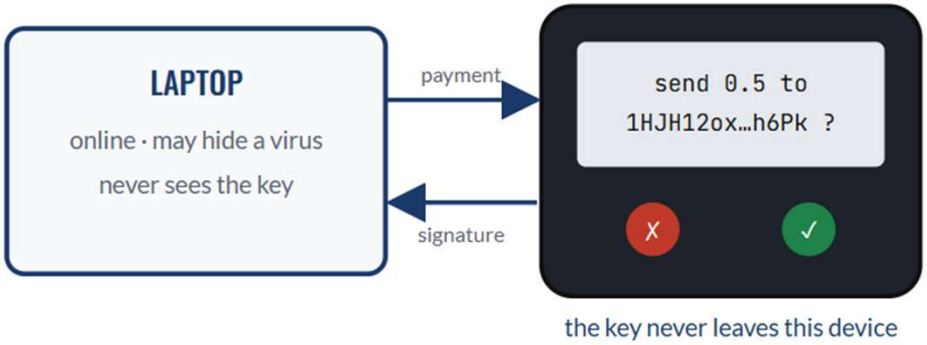
It protects the key, not your judgement

The device signs whatever you approve. A thief who tricks you into approving the wrong payment needs no key. Read the device's screen, not the laptop's.

My video

Lecture 4: a signature is computed from the message and the key. The key can stay inside the device; only the signature has to travel.

A hardware wallet signs inside itself: the key never touches the internet.







New Words: Custodial and Non-custodial

Lecture 1, again

A custodian is an intermediary. Blockchains were built to remove the intermediary — a custodial account quietly puts one back. Hence the saying "not your keys, not your coins".

My video

	Non-custodial wallet	Custodial wallet
Who holds the private key	you	a company — usually an exchange (where people buy and sell coins)
What you hold	the key, and the 12 words	a login and a password to the company's website
If you forget your password	Lecture 5: restore from the 12 words	the company can reset it for you
If the company fails	nothing changes	your coins are only a promise in its books
Example	your course wallet	an account at an exchange

New word · Custodial / non-custodial

Custodial: someone else holds the key for you. Non-custodial: you hold it yourself.

Custodial: someone else holds the key. Non-custodial: you do.

War Story: FTX, November 2022



FTX, then one of the largest exchanges in the world — a custodian for millions of customers

reports that its money was mixed with a sister company's; customers rushed to withdraw

withdrawals stopped; the company filed for bankruptcy

its founder was sentenced to 25 years in prison for fraud;
about 10 billion dollars taken from customers, investors and lenders, as reported

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Respect

Ordinary people lost savings they had trusted to a company that looked safe. Years of court work followed.

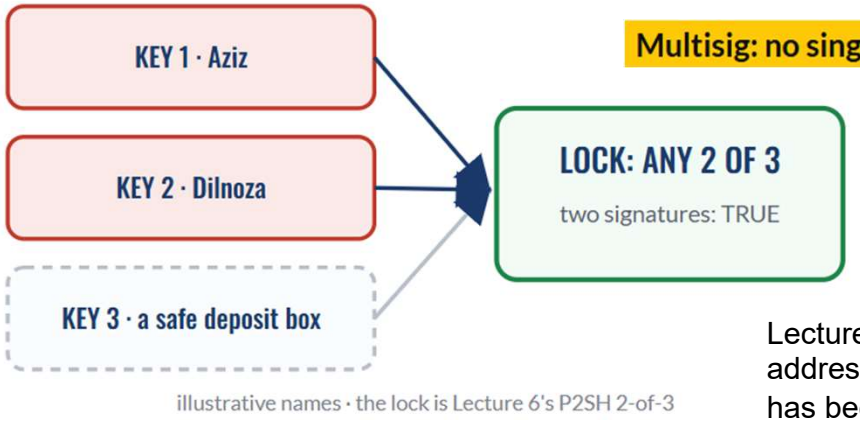
The engineering lesson

No blockchain failed. Bitcoin and Ethereum kept making blocks every minute. What failed was an intermediary — exactly the risk Lecture 1 described. A custodial balance is a promise, not a coin.

The chain did not fail; the custodian did.

New Word: Multisig

Multisig: no single key can move the coins.



illustrative names · the lock is Lecture 6's P2SH 2-of-3

New Word: Multisig - A lock that needs signatures from several keys: any m of n. Lecture 6's P2SH lock "any 2 of 3" is one.

Lecture 6's real 2-of-3 address `3CTkLec3…NDfJLG` has been used in **4,134** transactions so far — a company treasury or a service, most likely.

CALCULATE: How Many Keys Can You Lose?

Why 2 of 3 is the classic

Lose one key: still safe and usable. One key stolen: the thief cannot move anything. 1 of 1 — a normal wallet — fails at both.

Your turn

A family keeps savings in a 3-of-5 lock. Three keys are at home and the house burns down. Can they still spend? Can a burglar who took the three home keys?

Lock	Keys you may lose	Keys a thief must steal
1 of 1	0	1
2 of 2	0	2
2 of 3	1	2
3 of 5	2	3

CALCULATE An m-of-n lock: any m of the n keys must sign.

Keys you may lose = n − m; keys a thief needs = m.

My video

m of n: you may lose n − m keys; a thief needs m.

How do you change rules that nobody owns?

Two Questions Gone

Stop	Today's question	What we know	
The place	Where should a key live?	hot or cold; with you (non-custodial) or with a company (custodial); a hardware wallet signs inside itself	✅
The keys	Can one key be too few?	multisig: any m of n keys must sign	✅
The rules	How do the rules of a blockchain change?	—	❓
The split	What happens to my coins when a chain splits?	—	❓
The members	Who may read, write and join?	—	❓
The agreement	Can a network agree without mining?	—	❓

Two questions gone. Next: the rules.

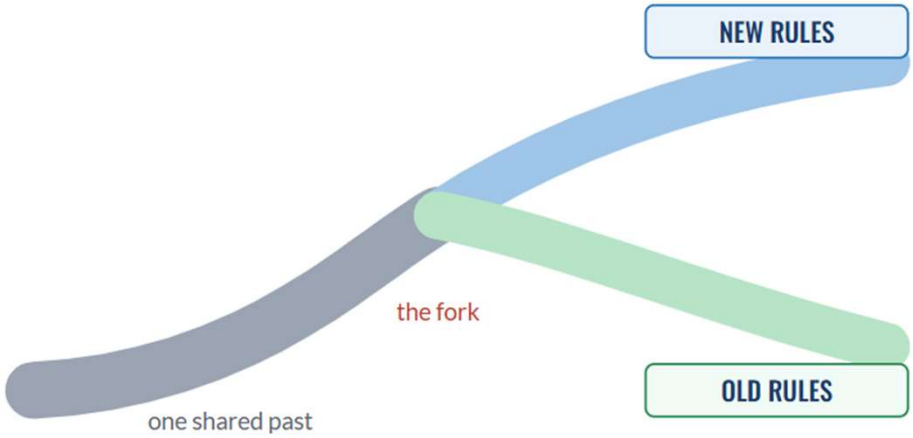
A Story: Changing the Rules of a Game

Imagine football with no federation: thousands of referees, each with a printed rulebook, all over the world. Someone publishes a new edition. Some referees switch; some keep the old one.

1. **A stricter rule** — "no more than 5 substitutes" instead of 7.
   A game played under the new rule is still a legal game to an old referee: 5 is allowed under both.
2. **A looser rule** — "goals from your own half count double".
   An old referee refuses that score. Old and new referees now disagree about who won.

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Tightening a rule keeps one game; loosening it can make two.



New Word: Fork

> 📘 **New word · Fork**
>
> A change of a blockchain's rules, which every node must choose to accept or not — named after a fork in a road, where one road becomes two.

A fork is a change of the rules; the nodes vote by what they run.

Nobody can force a node to update. The rules are whatever the nodes run. A change works only if enough people choose to run it.

Who writes the new rules? Usually volunteers, who publish a numbered proposal (a BIP for Bitcoin, an EIP for Ethereum — Lecture 5) and argue for it in public.

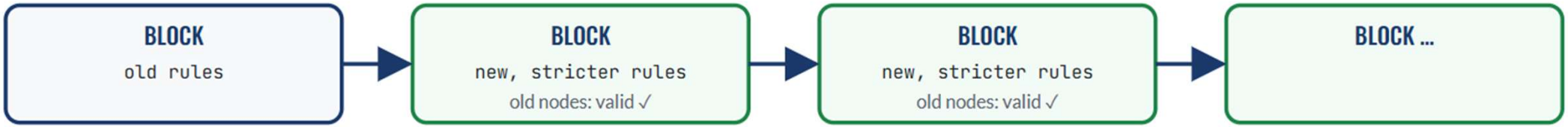
New Word: Soft Fork

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New word · Soft fork

A change that only makes the rules stricter. Blocks made under the new rules are still valid under the old ones, so nodes that did not update stay on the same chain.

ONE CHAIN · old and new nodes still agree



new blocks obey the old rules too — only more strictly — so nodes that did not update still accept them

You have met one

Lecture 6: SegWit, switched on at block 481,824 on 24 August 2017. Old nodes still accepted SegWit blocks — they simply could not check the new part.

A soft fork tightens the rules; one chain remains.

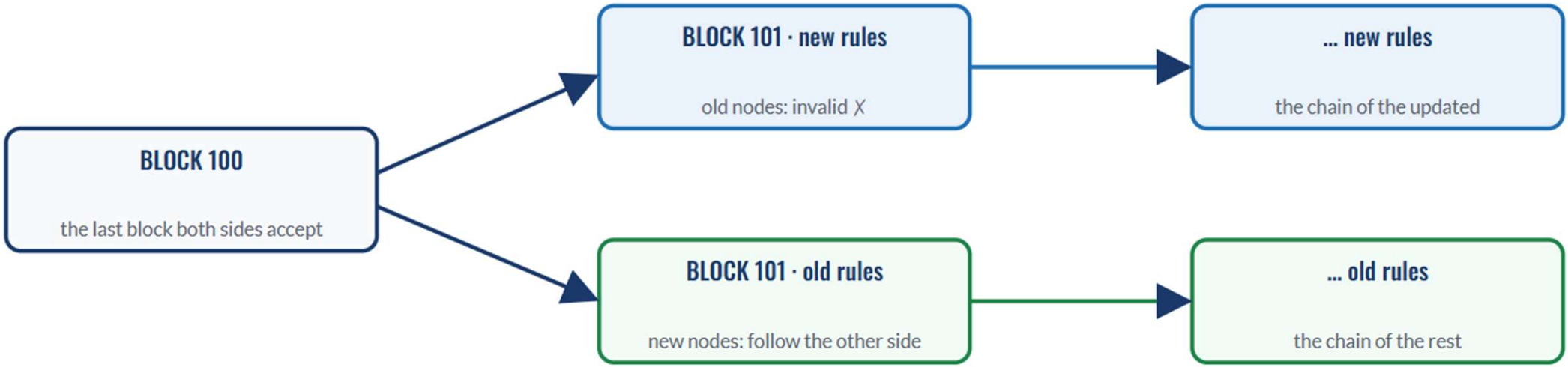
New Word: Hard Fork

New word · Hard fork

A change that loosens the rules or changes them outright. New blocks are invalid under the old rules, so everyone must update — or the chain splits into two.

If everyone updates, a hard fork is just an upgrade: Ethereum has made many. If a large group refuses, both sides keep going, with one shared past and two futures.

My video



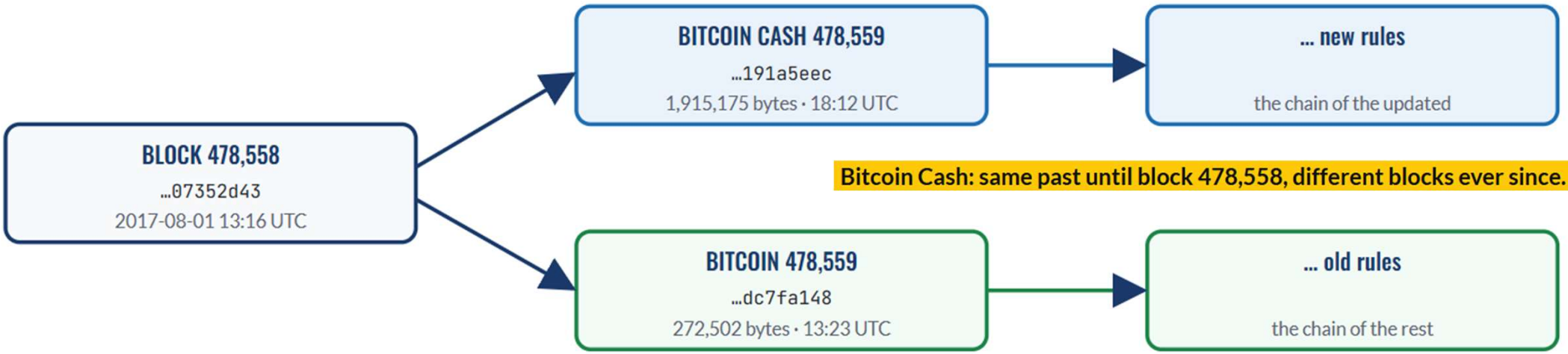
two chains with one shared past: every node follows the chain whose rules it runs

A hard fork loosens the rules; whoever does not follow ends up on another chain.

A Real Hard Fork: Bitcoin Cash, 1 August 2017

My video

Some people wanted bigger blocks for more payments.
On 1 August 2017 they ran new rules. Their block 478,559 was **1,915,175** bytes — illegal for Bitcoin, whose limit was 1,000,000.
Bitcoin's own 478,559 was 272,502 bytes.

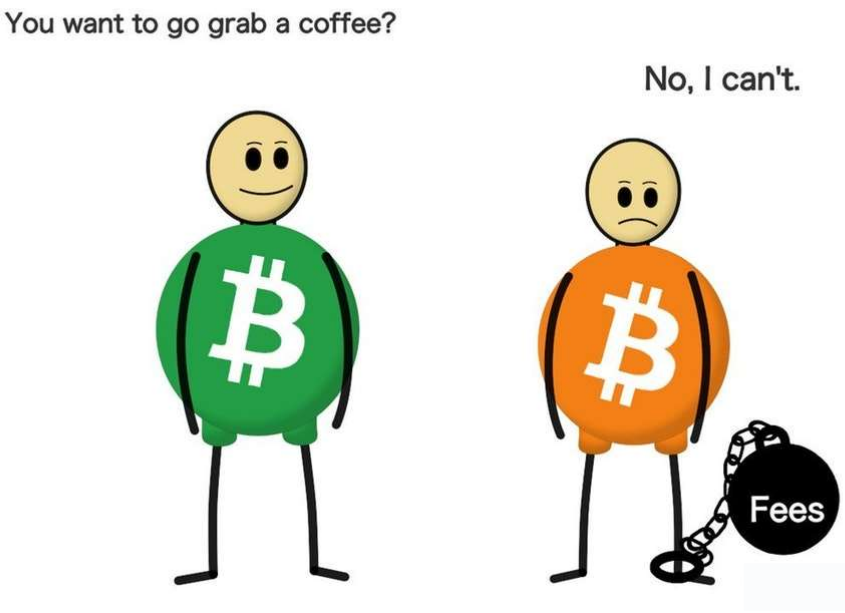


1 Aug 2017 · both build on the same block; the top one breaks the old 1,000,000-byte limit

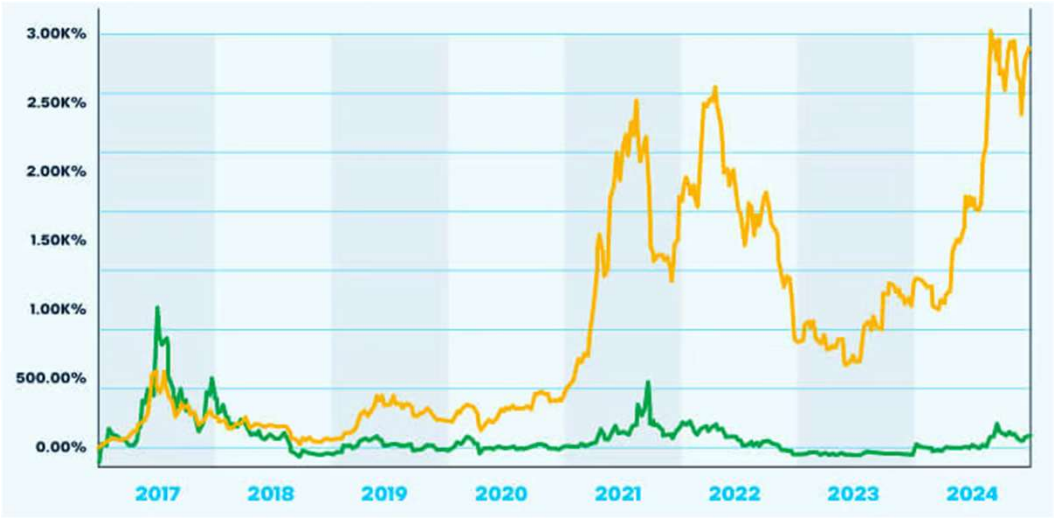
The new chain's first block came about **5 hours** after the last shared one:
few miners had followed, and blocks stayed as hard to make as before.

A Real Hard Fork: Bitcoin Cash vs Bitcoin

My video







Same Coins, Two Chains

At the moment of the split, every unspent output (Lecture 6) existed on **both** chains.
Whoever held a key for 1 BTC now also held 1 BCH, with the same key.

Your coins were…	After the split
in a non-custodial wallet	you held both: the same key opens the same outputs on each chain
at an exchange (custodial)	you received the new coins only if the exchange decided to give them to you

A split copies every coin; the key decides who owns both copies.

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The danger of one signature, two chains

If a payment's signature is valid on both chains, anyone can copy your payment from one chain to the other. Part 3 shows how this hurt Ethereum in 2016 — and the number Lecture 5 introduced that fixed it.

Two honest views

"Fix the theft: the rules exist to serve people." — "Code is law: if the rules can be changed after a loss, nothing is certain." Both sides kept their chain.

War Story: The DAO, June 2016

June 2016 DAO hack on the Ethereum network

What	The DAO — an investment fund that was a program on Ethereum (a smart contract, Weeks 9–11), holding money from thousands of people
17 Jun 2016	an attacker used a bug in the program to drain about 3.6 million ETH
The debate	change Ethereum's rules to give the money back — or accept the result, because the program allowed it?
20 Jul 2016	most of the community ran a hard fork at block 1,920,000 that moved the funds to a refund program

A program lost millions; the community had to choose between the money and the rules.

Block 1,920,000: Two Blocks, One Height

Shared block 1,919,999	`0xa218e2c611…807f2308`
Ethereum 1,920,000	`0x4985f5ca3d…059a11bb` · extra data `0x64616f2d68617264…` = "dao-hard-fork"
Ethereum Classic 1,920,000	`0x94365e3a8c…b28ade7f` · extra data "七彩神仙鱼", Chinese for a miner's name

Read the bytes yourself

`6461 6f2d …` in hexadecimal is ASCII for "dao-hard-fork": the forking miners wrote it into their blocks, like Satoshi's headline in block 0.

My video



20 Jul 2016 · both build on block 1,919,999 · we re-hashed both headers ourselves

One parent, two children: that is a hard fork, in two blocks.

One history until 2016; two networks ever since.

	Ethereum	Ethereum Classic
Chain ID	1	61
Newest block	26,249,679	25,586,453
Gaps between its last 7 blocks	12, 12, 12, 12, 12, 12 s	16, 8, 5, 41, 6, 7 s
Difficulty of the newest block	**0** — no longer mined (Part 4)	2.2×10^{15} — still mined, like Bitcoin
The DAO's money	returned by the fork	left where the attacker put it

My video

Ethereum and Ethereum Classic Today

Same history until July 2016, two networks ever since — each with its own block writers, its own coin and its own explorer.

Ethereum's blocks come every 12 seconds like a clock; Ethereum Classic's come at random, the way mined blocks do.

📘 New word · Replay attack

Copying a payment signed for one chain and sending it on the other chain, where the same signature is valid too. After the 2016 split, people lost coins this way.

CALCULATE The chain ID hides in the signature's last number: v = 35 + 2 × chain ID + (0 or 1).

New Word: Replay Attack

```
Ethereum          v =  38 = 35 + 2 × 1 + 1
Ethereum Classic  v = 157 = 35 + 2 × 61 + 0
```

The chain ID inside every signature stops a payment from crossing to the other chain.

The fix (November 2016, Ethereum block 2,675,000): every signature now includes the **chain ID** — L5's number that "tells one network from another". A payment signed for chain 1 is invalid on chain 61.

Four Questions Gone

Keys and rules are explained. Two questions remain — and they reach beyond Bitcoin: must a blockchain be open to everyone, and must it be mined at all?

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Must a blockchain be open — and mined?

Stop	Today's question	What we know	
The place	Where should a key live?	hot or cold; with you (non-custodial) or with a company (custodial); a hardware wallet signs inside itself	✅
The keys	Can one key be too few?	multisig: any m of n keys must sign	✅
The rules	How do the rules of a blockchain change?	a soft fork makes them stricter, and old nodes still follow; a hard fork loosens them, and old nodes split off	✅
The split	What happens to my coins when a chain splits?	they exist on both chains; a chain ID in every signature stops a payment from being copied across	✅
The members	Who may read, write and join?	—	❓
The agreement	Can a network agree without mining?	—	❓

Four questions gone. Next: who may join, and how they agree.

New Words: Public and Private Blockchain

New word · Public blockchain

Anyone may read it, send transactions, run a node and take part in writing blocks — no permission needed. Bitcoin, Ethereum, Ethereum Classic.

New word · Private blockchain

One organisation decides who may read and write — for example a bank's internal ledger shared between its own branches.



My video

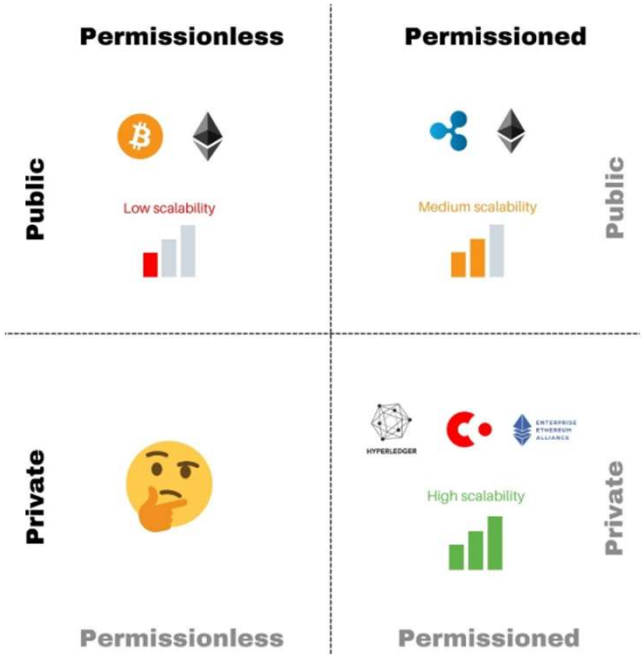
	Public	Private
Who may write	anyone who follows the rules	only computers the owner approves
Who may read	everyone, on any explorer	whoever the owner allows
Who can change the rules	whoever the nodes follow (forks)	the owner
Trust	in the rules and the work	in the owner — an intermediary again

Public: open to all. Private: one owner decides.

New Word: Permissioned Blockchain

📘 New word · Permissioned blockchain

A blockchain run by a known group of organisations — several banks, a group of shipping firms. Members are admitted; each runs nodes; no single one is in charge.



It sits between the other two: not open to everyone, not owned by one. Members know each other by name, so they do not need mining to keep each other honest — they need a way to vote.

My video

When does this make sense?

When several organisations that do not fully trust each other must keep one shared record, and the public must not see it.

Brochure vs reality

Many such projects were announced; fewer survived. When a simple shared database would do, it usually wins.

Permissioned: a known group, admitted by name, shares one ledger.

New Words: Proof of Stake and Validator

New word · Proof of stake

A way to choose who writes the next block without mining: people lock coins as a deposit — their **stake** — and are picked at random, more often the more they lock. Cheating destroys part of the deposit.

New word · Validator

A computer, with its locked coins, that takes part in proof of stake: it writes a block when chosen, and checks and votes on the blocks of others.



My video

Proof of stake replaces spent electricity with locked coins.

	Proof of work (Lecture 7)	Proof of stake
The cost of writing	electricity, spent guessing hashes	coins locked as a deposit
Who writes next	whoever guesses first	a validator picked at random, weighted by stake
Punishment for cheating	wasted electricity	part of the deposit is destroyed

The Day Ethereum Stopped Mining

	Block 15,537,393	Block 15,537,394
Time	2022-09-15 06:42:42 UTC	2022-09-15 06:42:59 UTC
Difficulty	11,055,787,484,078,698	0
Nonce	`0x62a3ee77461d4fc9`	`0x0000000000000000`
Written by	the last miner	the first validator

Seventeen seconds apart, the same chain switched from proof of work to proof of stake — without a split: everyone had updated. A hard fork that everyone followed.

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On 15 September 2022 Ethereum's difficulty fell to zero, for good.

📘 **New word · Finality**

The point after which a block can no longer be undone under the rules.

Not "immutable"

A blockchain is tamper-resistant: changing the past costs so much that it does not pay (Lecture 2). Finality is the rule that says when a block is past undoing — and different chains say it differently.

	Bitcoin (proof of work)	Ethereum (proof of stake)
When is a block final?	never completely — every block on top makes undoing harder (confirmations, Lecture 7)	after a two-thirds vote of the validators on a checkpoint: about **14 minutes** behind the newest block when measured
Undoing it would need	more work than the rest of the world, block after block	breaking a two-thirds vote, which destroys the voters' deposits

Finality: when a block is past undoing — slowly earned on Bitcoin, voted on Ethereum.

New Word: Finality

Most Popular Bitcion Wallets

ZenGo

#1 Best Overall

Coins Supported: ETH, BTC, DOGE, USDC, etc.
Type of wallet: Software wallet
Price: Free

crypto.com

Coins Supported: BTC, DOGE, DOT, ETH, LTC.
Type of wallet: Hot wallet
Price: Free

TREZOR

Coins Supported: BTC, ETH, LTC, DOGE, etc.
Type of wallet: Cold wallet
Price: $79

My video

Ledger Nano X

Coins Supported: BTC, ETH, LTC, DOGE, etc.
Type of wallet: Cold wallet
Price: $149

Coins Supported: 1000+ including BTC, ETH. **Type of wallet:** Hot wallet
Price: Free

Pionex

Coins Supported: 400+ including BTC, ETH, etc.
Type of wallet: Cold wallet
Price: Free

Ledger Nano X
Best Cold Wallet for those on a Budget

My video

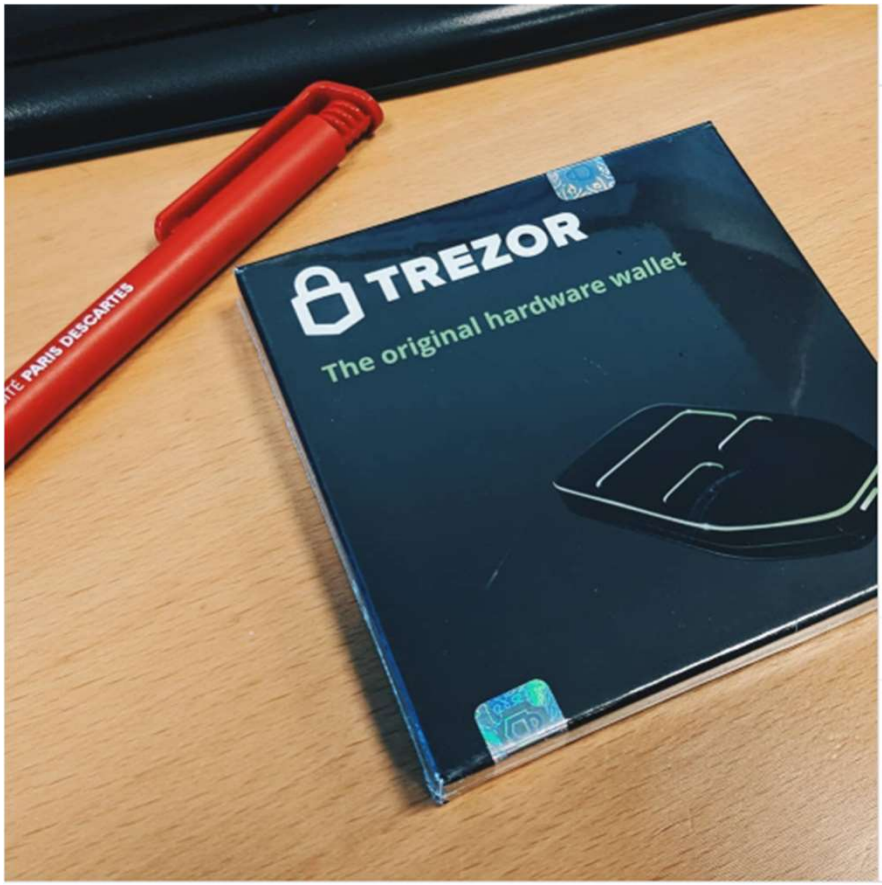


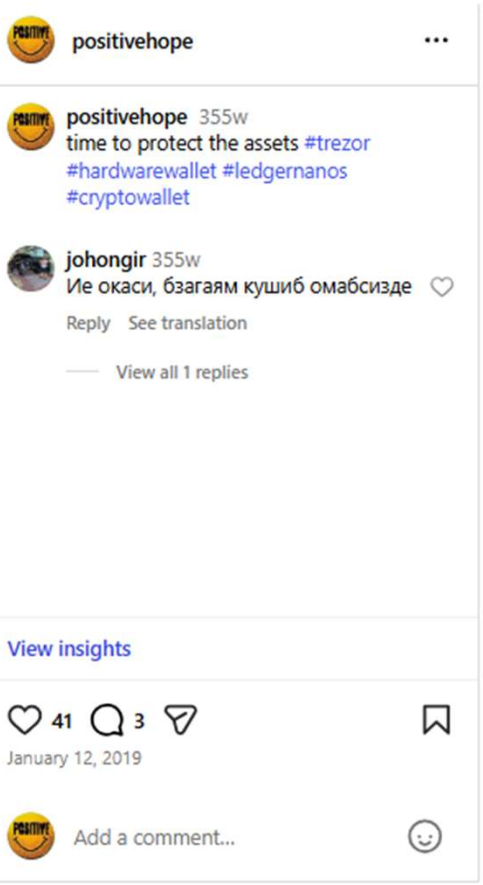
- Ledger Nano X is the advanced cold wallet by **Ledger** company.
- It supports over **100+ crypto** apps and coins.
- It is bundled with features like **Bluetooth** and USB-C cable connectivity. It allows you to easily manage the cryptos and **NFTs**.
- It strikes a good **balance** of accessibility and security.
- Ledger Nano X is packed with the most convenient form factors and uses **micro-LCD**.

- For the setup, you need to start with **setting up a PIN**, then a **24-word seed phrase**, which acts as your wallet's private key.
- It works best when paired with **Ledger live smartphone app**. Using the app, you can easily and quickly view your balance with or without the device.

Trezor – open source project

My video





…icy.

…t

…, **verify**.”

…cy hardware

…our private keys

Wallet Solutions

My video

gazeta.uz/ru/2025/10/31/asterium/

gazeta Политика Экономика Общество Культура Спорт Мир Колонки

Главная Новости Статьи Репортажи Медиа 4 ноября, вторник Afisha Spot Zi



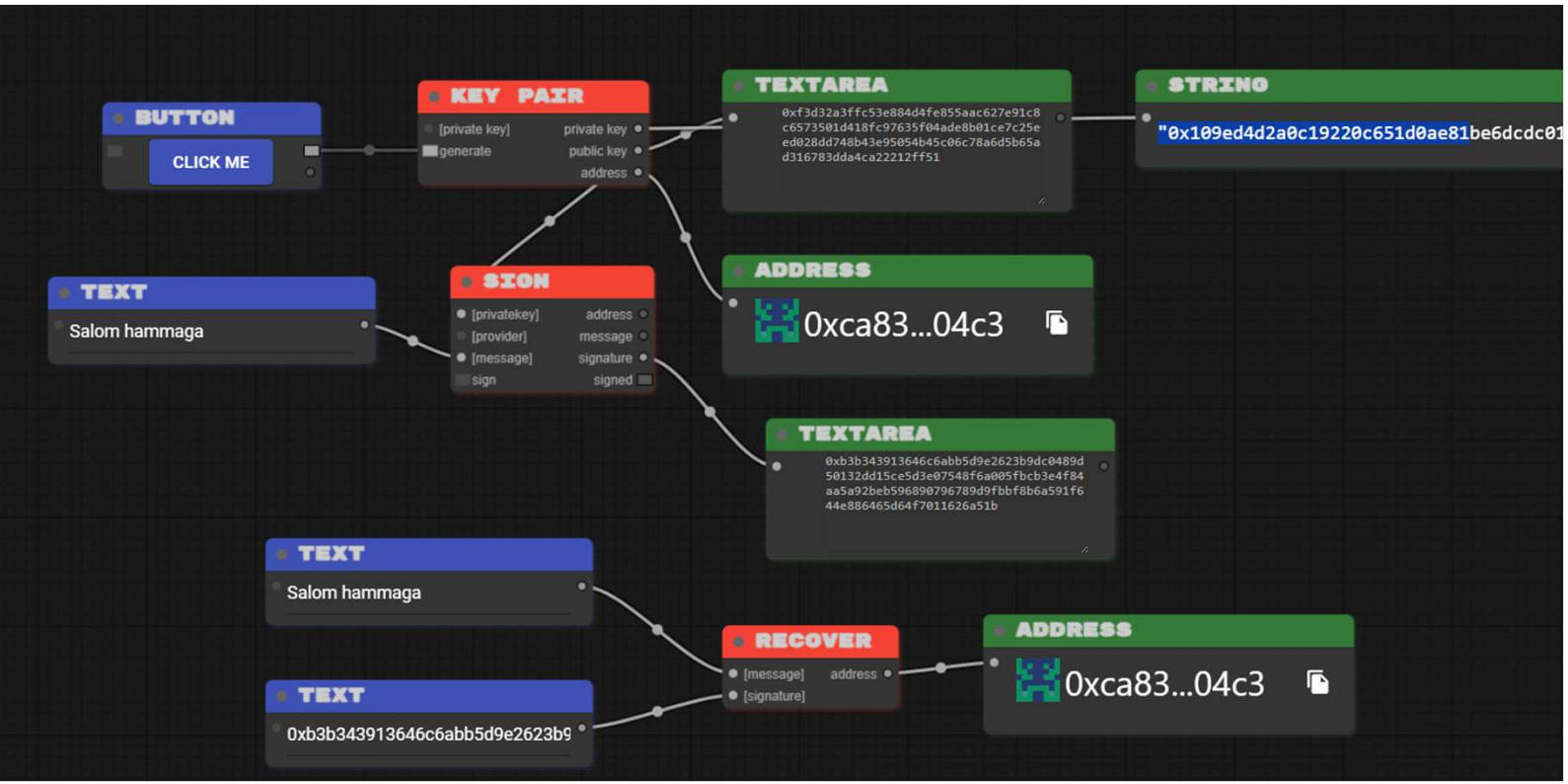
в Узбекистане

В Asterium представили три новых криптокарты, созданные совместно с Visa и HUMO. Они открывают пользователям возможность безопасно проводить онлайн- и офлайн-платежи с использованием цифровых активов.

Eth.build:

👉 Drag-and-Drop Programming

🧩 Open Source Building Blocks



An Educational Sandbox For Web3... And Much More.

My video

https://sandbox.eth.build



10/31/2026

SOC4190 2026 Yusupov